

Reading Summary

Language Model Scaling & Reinforcement Learning from Human Feedback (RLHF)

Internship Research Summary | Huzaifa Arshad | 2026

1. Language Model Scaling

Language Model Scaling refers to the systematic process of improving AI model performance by increasing three key resources: the number of parameters (model size), the volume of training data, and the amount of compute (processing power). Research has shown that increasing any of these dimensions leads to predictable improvements in model capability.

1.1 The Scaling Hypothesis

The scaling hypothesis, popularized by OpenAI and DeepMind researchers, proposes that larger models trained on more data consistently outperform smaller models. This was a major insight that shifted the field away from architectural tricks toward simply training bigger models.

1.2 The Three Pillars of Scaling

- **Parameters (Model Size):** The number of learnable weights in the neural network. GPT-3 has 175 billion parameters. More parameters = more capacity to learn complex patterns.
- **Training Data:** The amount of text (or other data) the model learns from. Models like GPT-4 were trained on trillions of tokens from the internet, books, and code.
- **Compute (FLOPs):** The total processing power used during training, measured in floating-point operations. More compute allows training larger models on more data.

1.3 Chinchilla Scaling Laws

In 2022, DeepMind published the Chinchilla paper, which was a turning point. It found that most large models at the time were actually undertrained — meaning they had too many parameters for the amount of data they were trained on.

The key insight: for a given compute budget, you should train a smaller model on more data rather than a larger model on less data. Chinchilla (70B params, 1.4T tokens) outperformed GPT-3 (175B params) despite being smaller, because it was trained on far more data.

1.4 Emergent Abilities

One fascinating result of scaling is emergent abilities — capabilities that suddenly appear at a certain model size that were completely absent in smaller models. Examples include multi-step reasoning, code generation, and few-shot learning. These abilities were not explicitly trained for; they emerged from scale alone.

2. Reinforcement Learning from Human Feedback (RLHF)

RLHF is the training technique that transforms a raw language model into a helpful, harmless, and honest AI assistant. It is the core method behind ChatGPT, Claude, and Gemini. Without RLHF, a large language model simply predicts the next token — it does not actually try to be helpful or safe.

2.1 The Problem RLHF Solves

Pre-trained language models are trained to predict text from the internet — which includes harmful, biased, and misleading content. They have no concept of "helpful" or "harmful." RLHF is a post-training technique that aligns the model with human values and preferences.

2.2 The Three Stages of RLHF

Stage 1: Supervised Fine-Tuning (SFT)

Human trainers write ideal responses to a set of prompts. The model is then fine-tuned on these examples to imitate the ideal behavior. This gives the model a starting point for being helpful.

Stage 2: Reward Model Training

The model generates multiple responses to prompts. Human raters rank these responses from best to worst. A separate "reward model" is trained to predict which responses humans prefer. This reward model acts as an automated judge of response quality.

Stage 3: Reinforcement Learning (PPO)

The main language model is then trained using Proximal Policy Optimization (PPO) — a reinforcement learning algorithm. The model generates responses, the reward model scores them, and the model is updated to generate higher-scoring responses over time. This loop continues until the model consistently produces responses that humans prefer.

2.3 Key Takeaway

RLHF is what makes the difference between a raw language model (which just predicts text) and a helpful assistant (which tries to answer your questions helpfully and safely). Scaling gives the model knowledge and ability; RLHF gives it alignment with human intent.

3. Quick Reference Summary

Concept	What It Is	Why It Matters
Scaling	Increasing model size, data, and compute	Bigger models = smarter models (up to a point)
Chinchilla Laws	Optimal balance of model size vs. data	Smaller model + more data can beat a giant model
Emergent Abilities	New capabilities that appear with scale	Models gain surprising skills at certain sizes
RLHF	Training with human preference feedback	Turns a raw model into a helpful assistant
Reward Model	AI that scores response quality	Automates human judgment at scale
PPO	Reinforcement learning algorithm	Optimizes the model to generate preferred outputs